
6.S183: A Practical Introduction to Diffusion Models

Distillation

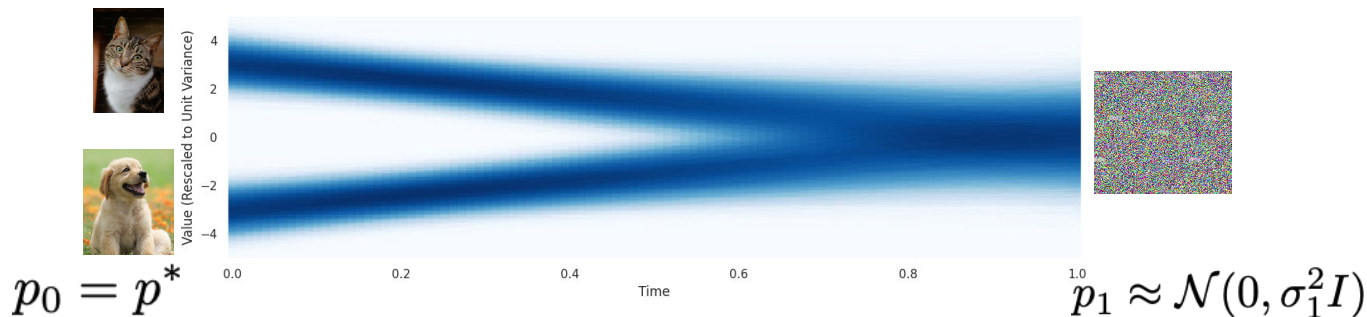
Artem Lukoianov
January 11, 2026



Recap: building probability path

Noising process: $x_t \stackrel{\text{dist}}{=} x_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$

→ Induces time-varying marginal $p_t(x_t)$ with $p_0 = p^*$ $p_1 \approx \mathcal{N}(0, \sigma_1^2 I)$



Recap: Deterministic vs Stochastic Sampling

Continuous ODE-Flow

$$\dot{x}_t = v(x_t, t)$$

$$v(x_t, t) = \dot{\sigma}(t) \mathbb{E}[\epsilon \mid x_t]$$

Continuous (Backwards) SDE

$$dx = f(x, t) \overleftarrow{dt} + \sqrt{2\rho(t)} \overleftarrow{dB}_t$$

$$f(x, t) = \left(\dot{\sigma}(t) - \frac{\rho(t)}{\sigma(t)} \right) \mathbb{E}[\epsilon \mid x_t]$$

Key point 1



Both methods traverse $p_t(x)$ when using $x_1 \sim \mathcal{N}(0, \sigma_1^2 I)$ approximation

Key point 2

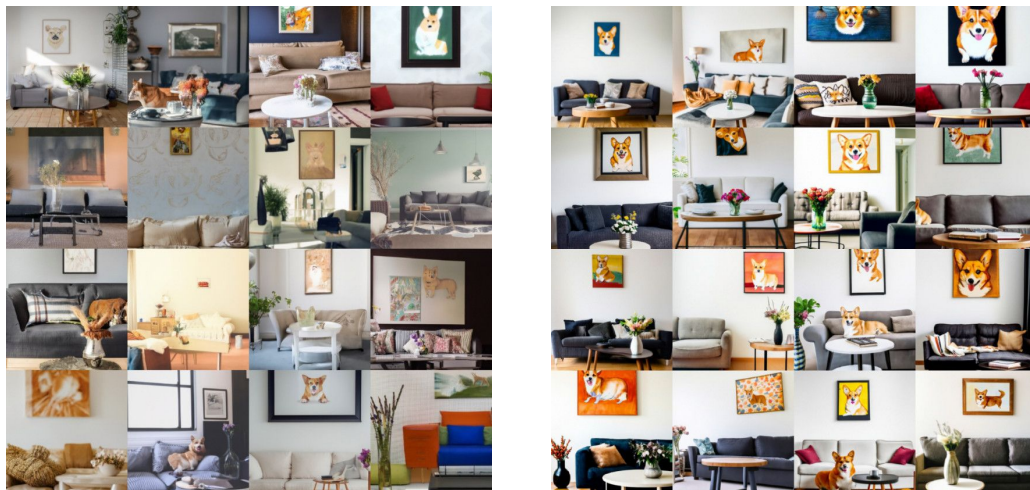


Having access to $\epsilon^*(x_t, t) = \mathbb{E}[\epsilon \mid x_t]$, for $t \in [0, 1]$ is sufficient to sample



Recap: Classifier-Free Guidance

$$\nabla_x \log p_t(x_t \mid y) = \underbrace{\nabla_x \log p_t(x_t)}_{\text{unconditional score}} + \underbrace{\nabla_x \log p_t(y \mid x_t)}_{\text{measurement-matching}}$$



OpenAI's GLIDE model, without guidance (left) and with guidance (right)

Today

What is distillation ?



What is distillation

- Generative models learn $p(x)$, but it is implicit
- We can distill $p(x)$ into
 - a smaller model
 - a fewer steps model
 - different modality
 - editing
 - anomaly detection

Outline

- What is distillation
- Few-step sampling
 - Rectified flows
 - Consistency models
 - Meanflow
- Score distillation
- Conclusion

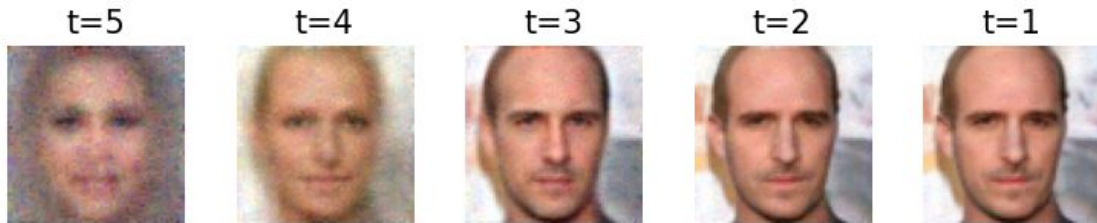
Outline

- What is distillation
- **Few-step sampling**
 - Rectified flows
 - Consistency models
 - Meanflow
- Score distillation
- Conclusion

Why multiple noise levels?

Generative models work best by iterative refinement

- LLMs autoregressively predict next token
- Diffusion models predicts increasingly fine details of image



Power law in natural image spectra

Magnitude of frequency-domain spectrum after applying 2D Fourier transform

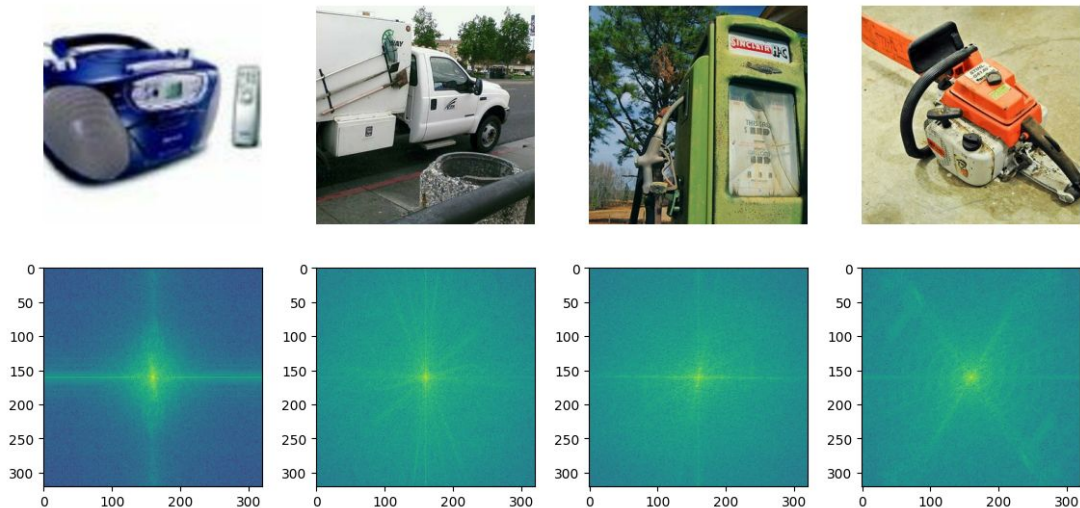


Image credit: Diffusion is spectral autoregression <https://sander.ai/2024/09/02/spectral-autoregression.html>



Power law in natural image spectra

Radially- averaged power spectral density (RAPSD)

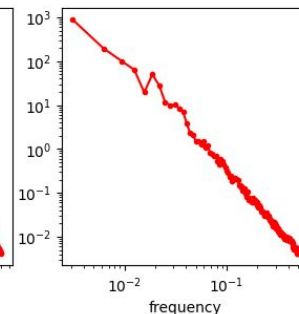
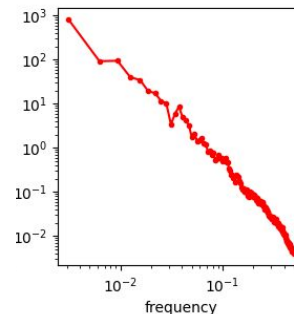
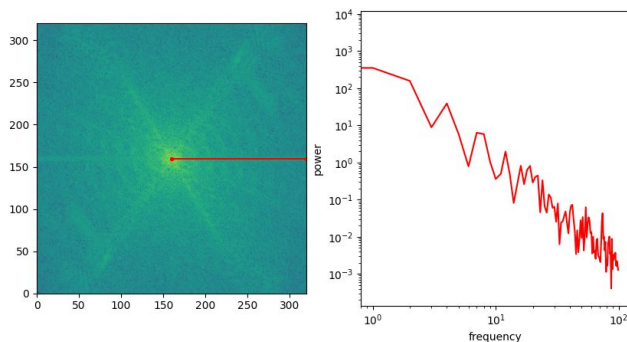


Image credit: Diffusion is spectral autoregression <https://sander.ai/2024/09/02/spectral-autoregression.html>



Power law in natural image spectra

Low frequencies dominate in magnitude/power compared to high frequencies (linear on log-log plot)

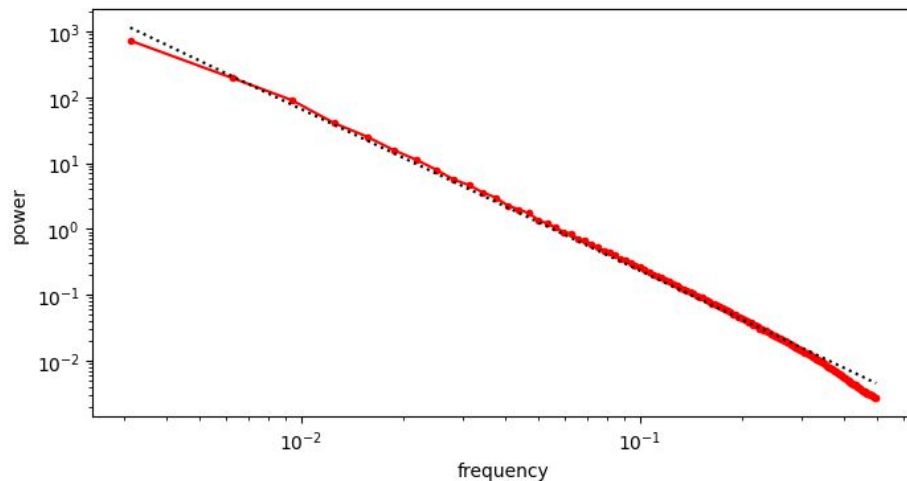
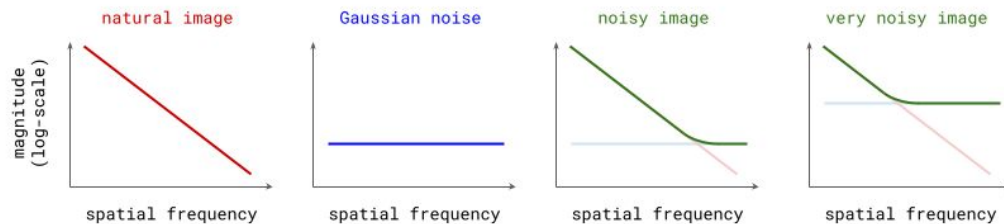


Image credit: Diffusion is spectral autoregression <https://sander.ai/2024/09/02/spectral-autoregression.html>

Gaussian noise first corrupts high freq. features

Gaussian noise has uniform power spectral density



Adding Gaussian noise “drowns out” high frequency features first

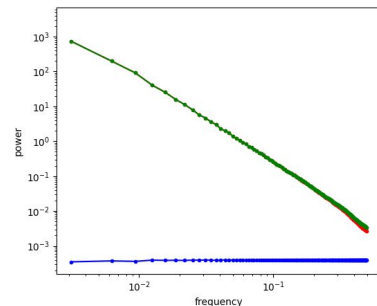
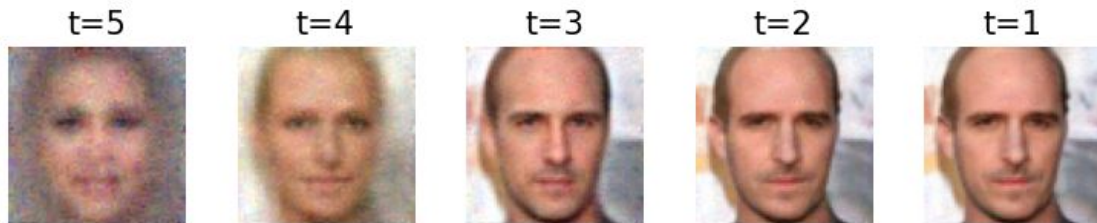


Image credit: Diffusion is spectral autoregression <https://sander.ai/2024/09/02/spectral-autoregression.html>

Denoising reconstructs low to high freq.

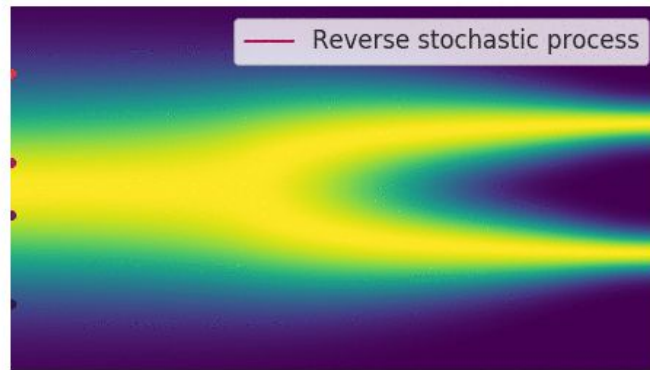
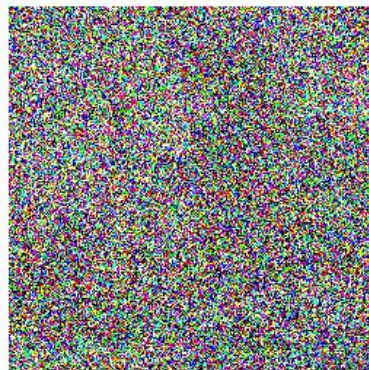
Initial denoising steps reconstruct basic shape (predicts an average of training data)

Final steps “fill in” finer details



In other words

$$v(x_t, t) = \dot{\sigma}(t) \mathbb{E} [\epsilon \mid x_t]$$



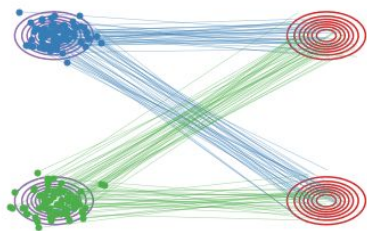
Reverse process. Image courtesy of Yang Song. <https://yang-song.net/>

How to sample faster?

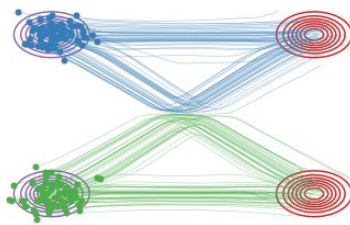
1. **Lecture 1:** Use fast deterministic samplers (training-free)
2. Additional training to speed up sampling
 - Distillation
 - Consistency models

Rectified flow

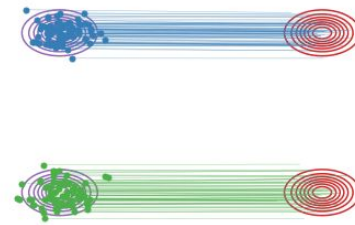
Image Credit: Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow Xingchao Liu, Chengyue Gong, Qiang Liu



Linear interpolation X_t



Rectified Flow Z_t



Straightened Rectified Flow

$$X_t = tX_1 + (1 - t)X_0, \quad t \in [0, 1].$$

$$v(z, t) = \mathbb{E}[X_1 - X_0 \mid X_t = z],$$

Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow

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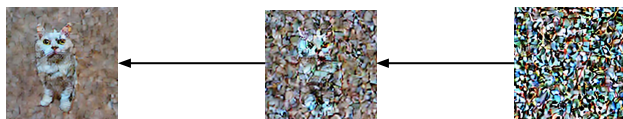
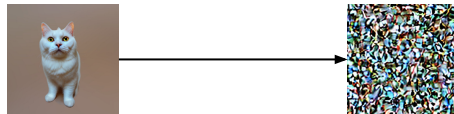
Abstract

We present rectified flow, a surprisingly simple approach to learning (neutral) ordinary differential equation (ODE) models to transport between two empirically observed distributions π_0 and π_1 , hence providing a unified solution to generative modeling and domain transfer, among various other tasks involving distribution transport. The idea of rectified flow is to learn the ODE to follow the straight paths connecting the points drawn from π_0 and π_1 as much as possible. This is achieved by solving a straightforward nonlinear least squares optimization problem, which can be easily scaled to large models without introducing extra parameters beyond standard supervised learning. The straight paths are special and preferred because they are the shortest paths between two points, and can be simulated exactly without time discretization and hence yield computationally efficient models. We show that the procedure of learning a rectified flow from data, called rectification, turns an arbitrary coupling of π_0 and π_1 to a new deterministic coupling with provably non-increasing convex transport costs. In addition, recursively applying rectification allows us to obtain a sequence of flows with increasingly straight paths, which can be simulated accurately with coarse time discretization in the inference phase. In empirical studies, we show that rectified flow performs superiorly on image generation, image-to-image translation, and domain adaptation. In particular, on image generation and translation, our method yields nearly straight flows that give high quality results even with a single Euler discretization step.



Progressive distillation

1. Train a teacher model
2. Sample clean image, add noise
3. Take 2 DDIM steps using teacher model



4. Train **student model** to predict 2 steps:
reduces total sampling steps by half

PROGRESSIVE DISTILLATION FOR FAST SAMPLING OF DIFFUSION MODELS

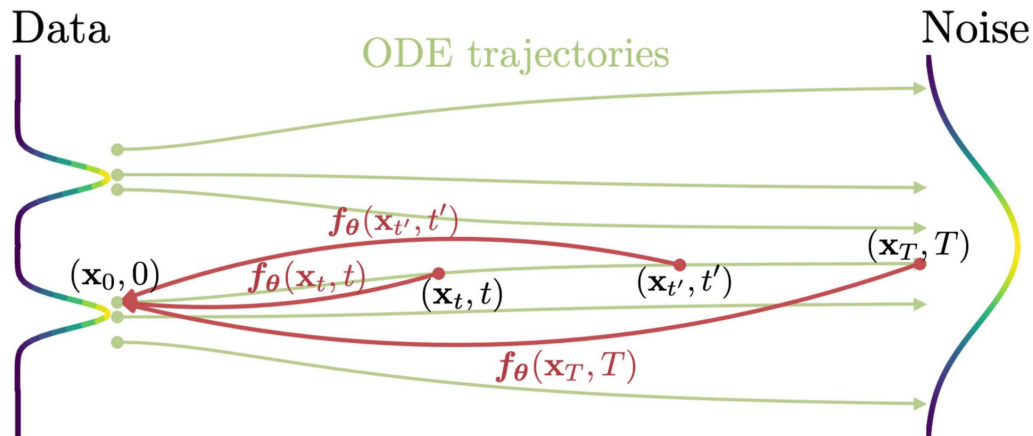
Tim Salimans & Jonathan Ho
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ABSTRACT

Diffusion models have recently shown great promise for generative modeling, outperforming GANs on perceptual quality and autoregressive models at density estimation. A remaining downside is their slow sampling time: generating high quality samples takes many hundreds or thousands of model evaluations. Here we make two contributions to help eliminate this downside: First, we present new parameterizations of diffusion models that provide increased stability when using few sampling steps. Second, we present a method to distill a trained deterministic diffusion sampler, using many steps, into a new diffusion model that takes half as many sampling steps. We then keep progressively applying this distillation procedure to our model, halving the number of required sampling steps each time. On standard image generation benchmarks like CIFAR-10, ImageNet, and LSUN, we start out with state-of-the-art samplers taking as many as 8192 steps, and are able to distill down to models taking as few as 4 steps without losing much perceptual quality; achieving, for example, a FID of 3.0 on CIFAR-10 in 4 steps. Finally, we show that the full progressive distillation procedure does not take more time than it takes to train the original model, thus representing an efficient solution for generative modeling using diffusion at both train and test time.



Consistency models



$$\mathcal{L}_{\text{CM}} = \|f_\theta(x_t, t) - \text{stopgrad}(f_\theta(x_s, s))\|_2^2$$

Consistency Models

Yang Song¹ Pradyumn Dharwal¹ Mark Chen¹ Ilya Sutskever¹

Abstract

Diffusion models have significantly advanced the fields of image, audio, and video generation, but they depend on an iterative sampling process that causes slow generation. To overcome this limitation, we propose *consistency models*, a new family of models that generate high quality samples by directly mapping noise to data. They support fast one-step generation by design, while still allowing multistep sampling to trade compute for sample quality. They also support zero-shot data editing, such as image inpainting, colorization, and super-resolution, without requiring explicit training on these tasks. Consistency models can be trained either by distilling pre-trained diffusion models, or as standalone generative models altogether. Through extensive experiments, we demonstrate that they outperform existing distillation techniques for diffusion models in one- and few-step sampling, achieving the new state-of-the-art FID of 3.55 on CIFAR-10 and 6.20 on ImageNet 64 × 64 for one-step generation. When trained in isolation, consistency models become a new family of generative models that can outperform existing one-step, non-adversarial generative models on standard benchmarks such as CIFAR-10, ImageNet 64 × 64 and LSUN 256 × 256.

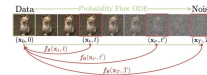
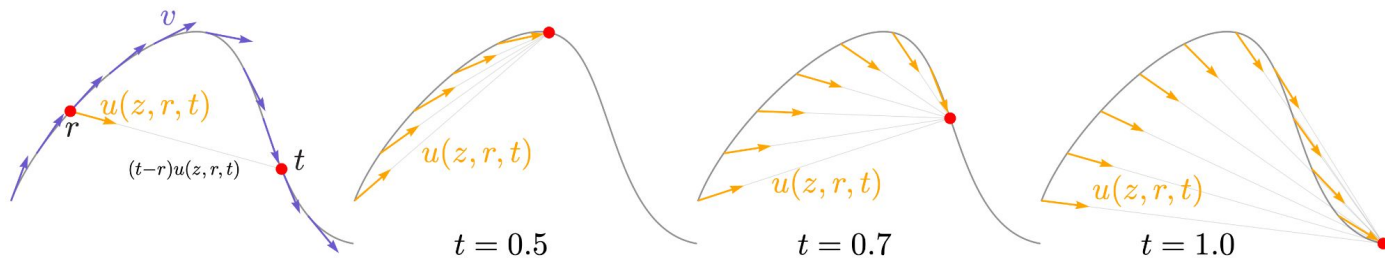


Figure 1: Given a *Probability Flow (PF) ODE* that smoothly converts data to noise, we learn to map any point (e.g., $\mathbf{x}_0$, $\mathbf{x}_t$, and $\mathbf{x}_T$) on the ODE trajectory to its origin (e.g., $\mathbf{x}_0$) for generative modeling. Models of these mappings are called *consistency models*, as their outputs are trained to be consistent for points on the same trajectory.

2022b). A key feature of diffusion models is the iterative sampling process which progressively removes noise from random initial vectors. This iterative process provides a flexible trade-off of compute and sample quality, as using extra compute for more iterations usually yields samples of better quality. It is also the crux of many zero-shot data editing capabilities of diffusion models, enabling them to solve challenging inverse problems ranging from image inpainting, colorization, stroke-guided image editing, to Computed Tomography and Magnetic Resonance Imaging (Song & Ermon, 2019; Song et al., 2021, 2022, 2023; Kawar et al., 2021, 2022; Chung et al., 2023; Meng et al., 2021). However, compared to single-step generative models like



Meanflow



$$u(z_t, r, t) \triangleq \frac{1}{t-r} \int_r^t v(z_\tau, \tau) d\tau.$$

$$\underbrace{u(z_t, r, t)}_{\text{average vel.}} = \underbrace{v(z_t, t)}_{\text{instant. vel.}} - (t-r) \underbrace{\frac{d}{dt} u(z_t, r, t)}_{\text{time derivative}}$$

$$\frac{d}{dt} u(z_t, r, t) = v(z_t, t) \partial_z u + \partial_t u,$$

Mean Flows for One-step Generative Modeling

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¹CMU ²MIT

Abstract

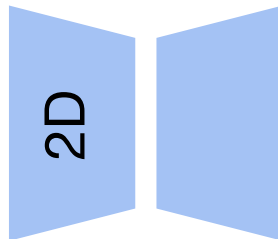
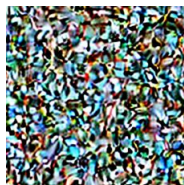
We propose a principled and effective framework for one-step generative modeling. We introduce the notion of average velocity to characterize flow fields, in contrast to instantaneous velocity modeled by Flow Matching methods. A well-defined identity between average and instantaneous velocities is derived and used to guide neural network training. Our method, termed the *MeanFlow* model, is self-contained and requires no pre-training, distillation, or curriculum learning. MeanFlow demonstrates strong empirical performance: it achieves an FID of 3.43 with a single function evaluation (1-NFE) on ImageNet 256×256 trained from scratch, significantly outperforming previous state-of-the-art one-step diffusion/flow models. Our study substantially narrows the gap between one-step diffusion/flow models and their multi-step predecessors, and we hope it will motivate future research to revisit the foundations of these powerful models.



Outline

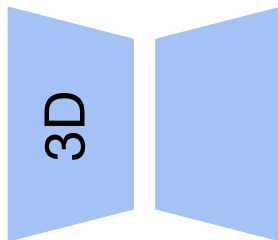
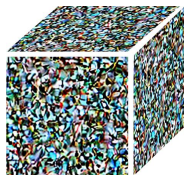
- What is distillation
- Few-step sampling
- **Score distillation**
- Conclusion

What do we want to do ?



$\sim 10^9$

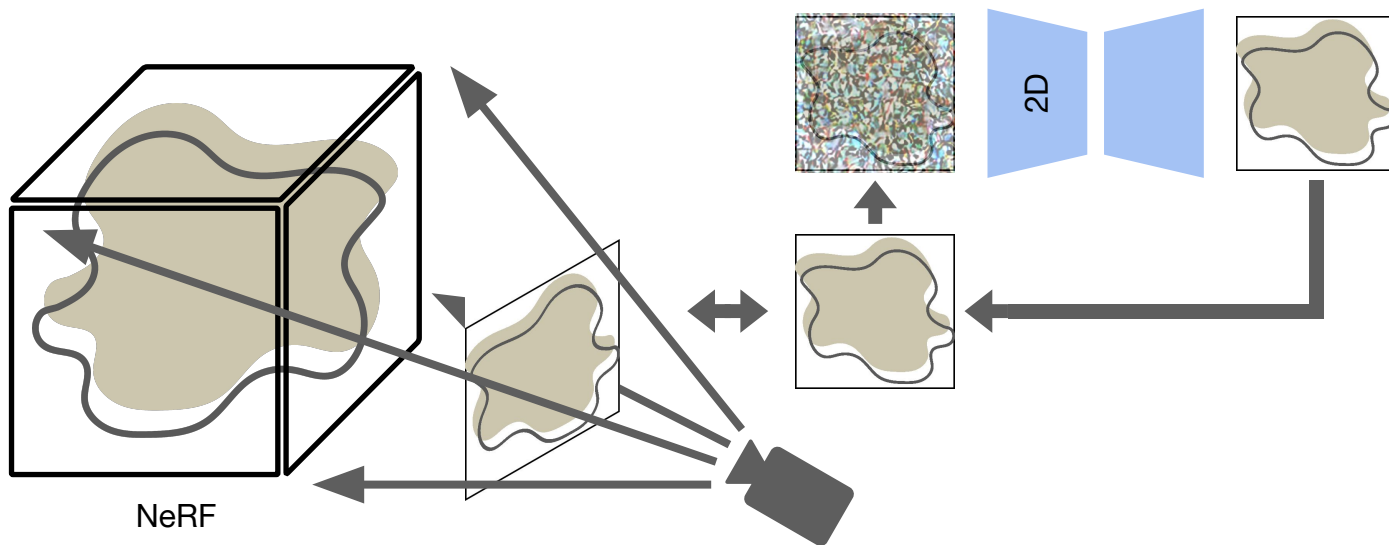
"A photograph of a
cat"



$\sim 10^5$

"A 3D model of a
cat"

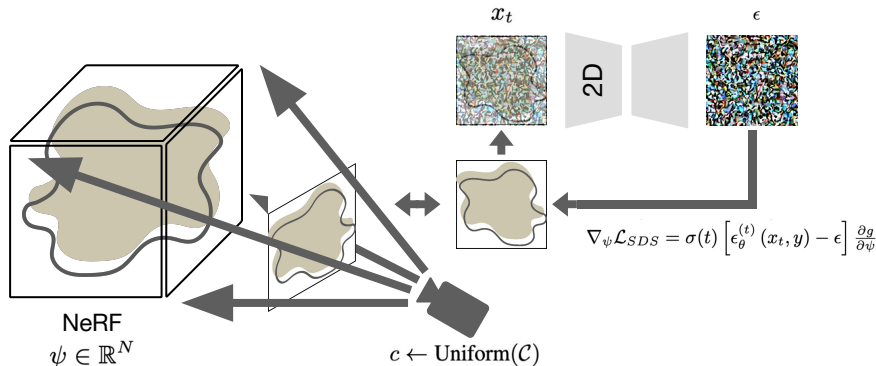
Score Distillation Sampling ?



Poole, Ben, et al. "Dreamfusion: Text-to-3d using 2d diffusion." *arXiv preprint arXiv:2209.14988* (2022).

Wang, Haochen, et al. "Score jacobian chaining: Lifting pretrained 2d diffusion models for 3d generation." *Proceedings of the IEEE/CVF CVPR*. 2023.

How does Score Distillation Work ?



$$\nabla_{\psi} \mathcal{L}_{SDS} = \sigma(t) \left[\epsilon_\theta^{(t)}(x_t, y) - \epsilon \right] \frac{\partial g}{\partial \psi}$$

Algorithm 1 Dreamfusion (SDS)

Input: $\psi \in \mathbb{R}^N$ - parametrized 3D shape
 $\mathcal{C}$ - set of cameras around the 3D shape
 y - text prompt
 $g : \mathbb{R}^N \times \mathcal{C} \rightarrow \mathbb{R}^{n \times n}$ - differentiable renderer
 $\epsilon_\theta^{(t)} : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$ - trained diffusion model

Output: 3D shape ψ of y

procedure DREAMFUSION(y)

for i in range(n_iters) **do**

$t \leftarrow \text{Uniform}(0, 1)$

$c \leftarrow \text{Uniform}(\mathcal{C})$

$\epsilon \leftarrow \text{Normal}(0, I)$

$x_t \leftarrow \sqrt{\alpha(t)}g(\psi, c) + \sqrt{1 - \alpha(t)}\epsilon$

$\nabla_{\psi} \mathcal{L}_{SDS} = \sigma(t) \left[\epsilon_\theta^{(t)}(x_t, y) - \epsilon \right] \frac{\partial g}{\partial \psi}$

 Backpropagate $\nabla_{\psi} \mathcal{L}_{SDS}$

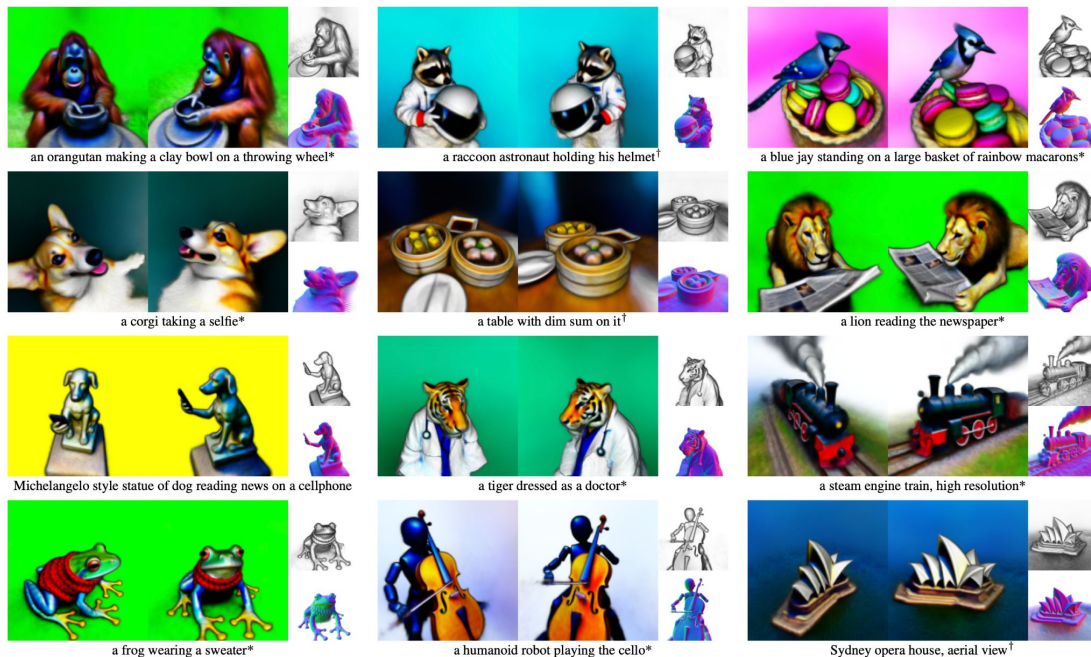
 SGD update on ψ

Poole, Ben, et al. "Dreamfusion: Text-to-3d using 2d diffusion." *arXiv preprint arXiv:2209.14988* (2022).

Wang, Haochen, et al. "Score jacobian chaining: Lifting pretrained 2d diffusion models for 3d generation." *Proceedings of the IEEE/CVF CVPR*. 2023.

Results

SDS = Score Distillation Sampling



Poole, Ben, et al. "Dreamfusion: Text-to-3d using 2d diffusion." *arXiv preprint arXiv:2209.14988* (2022).

ProlificDreamer: High-Resolution Video Generation with

Score distillation sampling (SDS) by distilling pretrained large-over-saturation, over-smooth propose to model the 3D priors in SDS and present a variational framework in text-to-3D generation. We use poor samples with both SDS works well with various CFG and simultaneously improve CFG weight (i.e., 7.5). We use space for text-to-3D such as which are orthogonal to the overall approach, dubbed *P*tion (i.e., 512×512) and effects (e.g., smoke and drop) by VSD are meticulously discussed at <https://ml.cs.tsinghua.edu.cn/~wcub/>

Score Distillation Sampling to-content generation in non-process and introduce a strategy for large Classifier-Free undesired noise term. Building Free Score Distillation (NFS) the original SDS framework effective distillation of pre-nominal CFG scale. This strategy of results, ensuring that the desired prompt. To demonstrate samples that compare NFSD

Image synthesis has recently witnessed a surge in research activity (Yu et al., 2022; Ding et al., 2022; Chang et al., 2023). Specifically, text-to-image methods leading the way (Sohl-Dickstein et al., 2016; Rombach et al., 2022; Balaji et al., 2022), followed by diffusion-based text-to-image models (Sohl-Dickstein et al., 2016; Rombach et al., 2022; Balaji et al., 2022). Sampling (SDS) was introduced. In the SDS loss which utilizes the strength of a NeRF (Mildenhall et al., 2020) as a mechanism can also be used to optimize texture maps (Metzger et al., 2022; Tsai et al., 2023), and SVG (Jain et al., 2023).

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<https://sds>

Score distillation sampling (SDS) [15] is the use of large-scale diffusion priors to improve the use of smaller-scale diffusion priors. Unfortunately, SDS has a number of weaknesses in general-purpose applications: (1) understanding the behavior of SDS requires an optimal-cost transport path from source to target. Under this new interpretation, these artifacts (source) to the natural image distribution are (1) characteristic artifacts are caused by (2) and (2) poor estimates of the source distribution. Conditioning of the source distribution results with little extra computation across many domains, matching or better than paintings to real images, optical illusion, compare our method to existing approaches that show that it can produce high-frequency

Diffusion models have shown tremendous results in image generation [49, 52, 3, 22], videos [57, 4] and audio [58]. However, they are often criticized for being slow and uninterpretable, they produce state-of-the-art results but at a high computational cost. One of the main reasons for this is the lack of sufficient training data. To overcome this, recent works have proposed diffusion models trained on domains with rich data, such as Score Distillation Sampling (SDS) [46, 67] and Image-to-Image Distillation [68], to optimize parametric images, *i.e.*, images produced by a parametric model. Despite being applicable to a wide range of domains, these models still suffer from several significant artifacts, such as oversaturation and blurry details. To address these issues, we have proposed to alleviate these artifacts [68], *i.e.*, artifacts.

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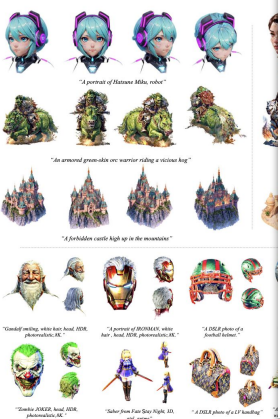


Figure 1. **Examples of text-to-3D content creations with our framework.** We present a text-to-3D generation framework, named the *LucidDreamer*, to distill high-fidelity textures and shapes from pretrained 2D diffusion models (detailed shows on Sec. 4) with a novel **Interval Score Matching** objective and an *Advanced 3D distillation pipeline*. Together, we achieve superior 3D generation results with photorealistic quality in a short training time. Please zoom in for details.

¹Massachusetts Institute of Technology ²University of Oxford ³MIT-IBM Watson AI Lab, IBM Research ⁴Toyota Research Institute ⁵Meta Reality Labs Research
<https://lukoianov.com/sdi/>

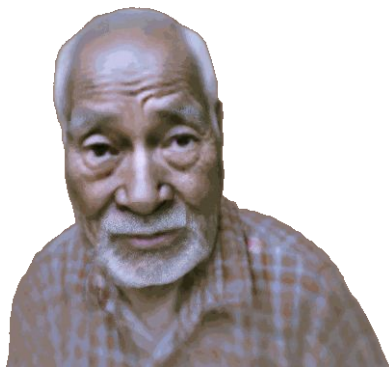
While 2D diffusion models generate realistic, high-detail images, 3D shape generation methods like Score Distillation Sampling (SDS) based on these 2D shape models produce cartoon-like, over-smoothed shapes. To help explain this discrepancy, we show that the image guidance used in Score Distillation can be interpreted as a stochastic differential equation (SDE) denoising process with the choice of a noise term. In particular, after a change of variables, SDS resembles a high-variance version of Denoising Diffusion Implicit Models (DDIM) with a differently-sampled noise term. SDS introduces noise i.i.d. randomly at each step, which can lead to over-smoothing and unrealistic outputs. We show that a better noise approximation can be recovered by inverting DDIM in each SDS update step. This modification makes SDS's generative process for 2D images almost identical to DDIM, and we show that this modification also improves the quality of 3D shapes and brings the generation quality closer to that of 2D samplers. Experimentally, our method achieves better or similar 3D generation quality compared to other state-of-the-art Score Distillation methods, all without training additional neural networks. Our work provides a new perspective on the relationship between 2D and 3D asset generation with diffusion models.



a) DDIM (CFG=7.5) b) 2D SDS (CFG=7.5) c) 2D ours (CFG = 7.5) d) 3D SDS (CFG=100) e) 3D ours (CFG=7.5)

We present a text-to-3D generation framework, named the 3D Diffusion Models (detailed shows on Sec. 4) with a novel architecture. Together, we achieve superior 3D generation results with

Better noise sample – better results



Takeaway from today

You can use a pre-trained image diffusion model and use it as a source of semantic gradients on images

That's all!

Feel free to reach out !

Feedback/Attendance Survey

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<https://forms.gle/7EJuu4euGxdFrUY37>